A Comparative Study of Machine Learning Models for Real-Time IoT- based Soil Moisture Anomaly Detection

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***Abstract*— Reliable real-time soil-moisture anomaly detection is critical for precision agriculture, yet performance can be compromised by class imbalance, noisy sensor data, and improper threshold selection. This study presents a controlled benchmarking analysis of seven machine learning models for binary soil-moisture anomaly detection using a publicly available agricultural dataset containing 100,000 samples. Anomalies were defined using a moisture-band labeling strategy, with the moisture feature excluded from predictors after label construction to prevent target leakage. The evaluated models include Support Vector Machine (RBF), k-Nearest Neighbors, Random Forest, XGBoost, LightGBM, CatBoost, and a deep Multilayer Perceptron. A stratified 70/15/15 train–validation–test split was adopted, with validation-based threshold optimization. Performance was assessed using F1-score, MCC, ROC-AUC, PR-AUC, and calibration metrics. Results indicate consistently strong performance across models, with F1-scores near 0.974 and PR-AUC approaching 0.999; LightGBM achieved the highest F1, while the Multilayer Perceptron and SVM demonstrated superior calibration. Limitations include reliance on a rule-based labeling scheme and evaluation on a single dataset, which may restrict generalizability.**

Keywords— IoT agriculture; soil moisture; anomaly detection; tabular learning; class imbalance; calibration.

I. Introduction

Precision agriculture deploys dense IoT sensing to monitor crop–soil interactions and to trigger timely interventions. Converting raw streams into reliable anomaly flags remains nontrivial because class priors are highly skewed, noise patterns drift over time, and cost trade-offs are asymmetric across false positives and false negatives [1], [2]. Simple accuracy averages can obscure failure modes under imbalance, while threshold choices that maximize one metric may degrade others. Probability miscalibration further distorts downstream decisions that rely on scores as risk proxies [3]–[5].

To reflect the families most commonly used for tabular sensing data, this benchmark includes margin-based kernel learning, instance-based learning, ensemble tree methods, modern gradient boosting, and neural networks. Support Vector Machines (SVMs) learn a maximum-margin decision boundary and use kernels (here, RBF) to model nonlinear separations in feature space [6]. k-Nearest Neighbors (KNN) is a nonparametric method that predicts from local neighborhoods and can be effective when class structure is locally clustered [7]. Random Forests aggregate many decorrelated decision trees via bagging and random feature subsampling, yielding strong baselines with robust generalization on heterogeneous tabular features [8]. Gradient boosting methods learn additive ensembles of trees; XGBoost introduces regularized, scalable tree boosting [9], LightGBM accelerates boosting with histogram-based split finding and leaf-wise growth [10], and CatBoost improves boosting robustness and categorical-feature handling via ordered boosting and related techniques [11]. Finally, multilayer perceptrons (MLPs) learn hierarchical nonlinear feature interactions through stacked affine transformations and nonlinear activations, providing a flexible parametric alternative when sufficient data are available [12]. These model classes are frequently reported across precision-agriculture analytics, motivating their inclusion for a controlled, leakage-safe comparison [1], [2].

An accuracy-focused, leakage-safe benchmark on a public dataset offers a controlled setting to study these effects. Leakage control is essential because target leakage inflates apparent performance and undermines external validity. A fixed train/validation/test split exposes models to identical difficulty, isolating the role of inductive bias rather than sampling luck. Validation-tuned thresholds approximate operational selection without contaminating the test set. Reporting both ranking metrics (ROC-AUC, PR-AUC) and operating-point metrics (Precision, Recall, F1) yields a more complete picture of separability and decision utility under skew [3]–[5].

This benchmark targets real-time soil-moisture anomaly detection from tabular features drawn from a widely used public dataset, enforcing leakage-safe preprocessing that uses Moisture only to derive the binary label, a single stratified 70/15/15 split for fair comparability, validation-tuned thresholds that maximize F1 before test reporting, and calibration plus robustness diagnostics to qualify headline results [1]–[5]. Together, these choices clarify metric behavior under skew and enable reproducible comparison of commonly deployed models in precision agriculture.

II. Related Work

Precision agriculture and crop-health monitoring increasingly integrate IoT sensors with AI models for prediction, anomaly detection, and decision support. Recent reviews highlight the rapid expansion of smart sensors and data-driven agricultural workflows, while emphasizing challenges including drift, missing data, and deployment constraints [1], [2]. In

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anomaly-centric settings, evaluation practice has evolved toward metric suites that separate ranking performance from operating-point behavior under skew; in particular, PR-AUC and MCC are often recommended alongside ROC-AUC to reduce misleading conclusions under imbalance [3], [5].

Modern reliability studies further show that high-capacity learners can produce over-confident probabilities; reliability diagrams and summary scores such as Expected Calibration Error (ECE) and Brier score are therefore recommended, alongside post-hoc methods for probability scaling [4]. These insights motivate reporting both discrimination and calibration, as calibrated scores enable cost-aware thresholding and stable alarm rates.

Building on these threads, the benchmark implements practices aligned with the literature by (i) strictly avoiding target leakage through careful feature curation and split hygiene, (ii) separating ranking ability (ROC-AUC, PR-AUC) from decision quality at a tuned operating point (Precision, Recall, F1, with MCC as a balance check), and (iii) assessing calibration with reliability plots and proper scoring rules to support threshold transfer across sites and seasons [1]–[5].

III. Dataset and Task Definition

The study uses a public Kaggle dataset referenced in [13], locally mirrored as *Plant_Parameters.csv*. The dataset contains 100,000 records and 10 columns: nine numeric features (pH, Soil EC, Phosphorus, Potassium, Urea, T.S.P, M.O.P, Moisture, Temperature) and one categorical feature (Plant Type) with 10 balanced classes [13].

The task is framed as binary anomaly detection. Samples whose volumetric water content falls within the moisture band [45, 55]% are labeled normal (0), whereas samples outside this band are labeled anomalous (1). To ensure leakage-free evaluation, Moisture is used only to derive the proxy label and is excluded from the predictor set. Under this proxy rule on this dataset, the resulting labels are strongly imbalanced (≈95% anomalous vs. ≈5% normal), which is preserved via stratified splitting [13]. For deployment, the "normal" band should be aligned with field capacity and permanent wilting point for the target soil texture and rooting depth, consistent with soil-physics and sensor-calibration guidance [14]–[18].

Domain context of the moisture band. The fixed "normal" moisture band is an operational proxy selected to enable a transparent and reproducible benchmark. In practice, feasible moisture ranges depend on soil texture, effective root-zone depth, sensor calibration, and local agronomic limits (e.g., field capacity and management targets). Accordingly, transferring this benchmark to other soils, crops, or climates may require re-deriving both the normal band and the alert threshold from local limits and sensor characteristics, rather than reusing the same numeric band unchanged [14]–[18].

IV. Methodology

Figure.1 illustrates the methodological workflow for real-time soil-moisture anomaly detection..The methodological pipeline begins with dataset acquisition followed by preprocessing, including cleaning, normalization, and feature selection. A rule-based moisture-band strategy is applied to generate anomaly labels, after which the moisture variable is removed to prevent target leakage. The dataset is then partitioned into stratified training, validation, and testing subsets (70/15/15). Seven machine learning models are trained and tuned using the validation set for threshold optimization. Final evaluation is conducted on the test set using classification, ranking, and calibration metrics. Statistical significance is assessed using bootstrap confidence intervals and McNemar tests, and models are compared to determine relative performance.

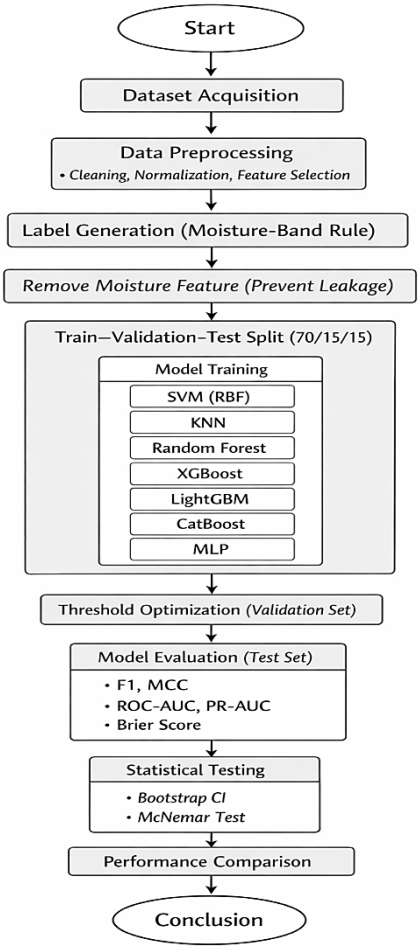


Figure.1 Methodological workflow for real-time soil-moisture anomaly detection

A. Problem Formulation

Let the dataset be defined as:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}, \quad x_i \in R^d,\ y_i \in \{0,1\}$$

where $x_i$represents the environmental and soil-related feature vector and $y_i$denotes the binary anomaly label (normal = 0, anomaly = 1).

The objective is to learn a function:

$$f_\theta : R^d \rightarrow [0,1]$$

that estimates the posterior probability:

$$P(y = 1 \mid x)$$

A classification decision is obtained using a validation-optimized threshold

$$\hat{y} = \begin{cases} \mathbf{1} & \text{if } f_\theta(x) \geq \tau \\ \mathbf{0} & \text{otherwise} \end{cases}$$

B. Applied Machine Learning Models

- **Support Vector Machine (RBF Kernel),** SVM seeks a maximum-margin hyperplane:

$$\min_{w,b,\xi} \quad \frac{1}{2}|w|^2 + C\sum_{i=1}^{N}\xi_i$$

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subject to:

$$y_i(w^\top \phi(x_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

where $\phi(\cdot)$is implicitly defined via the RBF kernel:

$$K(x_i, x_j) = \exp(-\gamma |x_i - x_j|^2)$$

- **k-Nearest Neighbors,** KNN classifies a sample based on the majority class among its $k$nearest neighbors:

$$\hat{y} = \text{mode}\{y_j : x_j \in \mathcal{N}_k(x)\}$$

where $\mathcal{N}_k(x)$ denotes the set of k nearest neighbors under Euclidean distance.

- **Gradient Boosting (XGBoost / LightGBM / CatBoost)**

Boosting models iteratively minimize a differentiable loss function.

$$\widehat{f^{(m)}}(x) = \widehat{f^{(m-1)}}(x) + \eta h_m(x)$$

where:

- $h_m(x)$is the weak learner (decision tree),
- $\eta$is the learning rate.

The objective typically includes regularization:

$$\mathcal{L} = \sum_{i=1}^{N} L(y_i, \hat{y_i}) + \Omega(h)$$

XGBoost applies second-order gradient optimization, LightGBM uses histogram-based splitting and leaf-wise growth, CatBoost employs ordered boosting to reduce prediction shift.

- **Multilayer Perceptron (MLP)**

The MLP models nonlinear transformations through layered mappings:

$$h^{(l)} = \sigma(W^{(l)} h^{(l-1)} + b^{(l)})$$

The output layer applies a sigmoid function:

$$\hat{y} = \sigma(W^{(L)} h^{(L-1)} + b^{(L)})$$

The model is trained by minimizing binary cross-entropy:

$$\mathcal{L} = -\sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Predictors are Temperature, pH, Soil EC, Phosphorus, Potassium, Urea, T.S.P, M.O.P, and encoded Plant Type, while Moisture is used only to derive the binary label. A fixed 70/15/15 train/validation/test split stratified on the label is applied, with standardization fit on the training split and applied to validation and test.

Seven baseline models are evaluated with standard settings: SVM (RBF, probability enabled), KNN (distance weighting), Random Forest (balanced class weights), XGBoost, LightGBM, CatBoost, and a Deep MLP with ReLU layers, dropout, L2 regularization, and early stopping. For each model, the decision threshold is tuned on the validation split to maximize F1 and then fixed on the test split.

Reported test metrics are Accuracy, Precision, Recall, F1, ROC-AUC, and PR-AUC, and MCC, Brier score, and ECE are computed for diagnostic analysis in line with recommendations for imbalanced detection and reliability assessment [3]–[5].

V. Experimental Setup

Seven supervised learning models, Support Vector Machine (RBF), k-Nearest Neighbors, Random Forest, XGBoost, LightGBM, CatBoost, and a shallow multilayer perceptron (DeepMLP) as a neural baseline were evaluated under a single, leakage-safe 70/15/15 stratified split (random seed = 42) to preserve class proportions. Numeric predictors were standardized using a StandardScaler fitted on the training set only, and the categorical *Plant Type* feature was label-encoded (encoding independent of the target label) and applied consistently across splits. The resulting transformations were applied unchanged to validation and test splits to avoid information leakage. For each model, the decision threshold was tuned on the validation set by maximizing the F1-score, after which the tuned threshold was fixed for all test reporting. Performance is summarized using threshold-dependent metrics (precision, recall, F1, MCC) and threshold-free metrics (ROC-AUC and PR-AUC). Probability reliability is assessed via Brier score and expected calibration error (ECE), following recommended reporting practice for calibrated decision-making under class skew [3]–[5]. Bootstrap confidence intervals and McNemar tests are used for robustness and pairwise disagreement assessment.

To facilitate reproducibility, the key hyperparameters and implementation settings for all evaluated models are summarized in Table 1.

Table 1. Key hyperparameters and implementation settings

Model	Key settings
XGBoost	objective=binary:logistic; n_estimators=800; max_depth=8; learning_rate=0.06; subsample=0.9; colsample_bytree=0.9; reg_lambda=1.0; eval_metric=logloss; tree_method=hist (gpu_hist if available); random_state=42
LightGBM	objective=binary; n_estimators=1000; num_leaves=63; learning_rate=0.06; subsample=0.9; colsample_bytree=0.9; random_state=42
CatBoost	loss_function=Logloss; iterations=1200; depth=8; learning_rate=0.06; random_seed=42
RandomForest	n_estimators=1000; max_depth=None; class_weight=balanced; random_state=42
SVM_RBF	kernel=rbf; C=2.0; gamma=scale; probability=True; class_weight=balanced; random_state=42
KNN	n_neighbors=31; weights=distance; metric=minkowski (p=2)
DeepMLP	4 hidden layers × 512 units (ReLU); dropout=0.3 after each hidden layer; L2=1e-4; output=sigmoid; optimizer=Adam (lr=1e-3); epochs⩽80; batch_size=512; EarlyStopping(patience=10, restore_best_weights=True)

VI. Results and Discussion

Table 2 reports the test performance under validation-tuned thresholds. All models form a tight utility cluster (F1 ≈ 0.974), indicating that under the chosen labeling policy and fixed split, discriminative performance is broadly similar across kernel, neighbor, boosting, bagging, and neural approaches.

Because the validation-tuned threshold is chosen to maximize F1, the resulting operating point can favor high-recall behavior under the observed prevalence; therefore, confusion-sensitive metrics (MCC) and calibration diagnostics are reported alongside F1 to better reflect decision quality for real-time alerting.

Table 2. Test performance with validation-tuned thresholds (test set)

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Model	Acc	Prec	Rec	F1	ROC-AUC	PR-AUC
LightGBM	0.9505	0.9541	0.9957	0.9745	0.9731	0.9986
DeepMLP	0.9501	0.9501	1.0000	0.9744	0.9746	0.9987
SVM_RBF	0.9501	0.9501	1.0000	0.9744	0.9739	0.9979
CatBoost	0.9501	0.9501	1.0000	0.9744	0.9729	0.9986
KNN	0.9500	0.9501	0.9999	0.9744	0.9740	0.9987
XGBoost	0.9497	0.9535	0.9956	0.9741	0.9731	0.9986
RandomForest	0.9491	0.9530	0.9956	0.9738	0.9729	0.9986

Impact of benchmark design choices. The near-identical F1 cluster observed in Table 2 should be interpreted as conditional on the operational labeling rule used in this benchmark (the fixed moisture band) and the resulting class proportions. Shifting or widening/narrowing the "normal" moisture band would change the prevalence of labeled anomalies and, consequently, the difficulty of the discrimination task and the operating-point trade-offs, potentially altering relative model rankings. Therefore, the reported metrics primarily characterize model behavior under this explicit proxy definition and are not claimed as universal agronomic ground truth. In this setting, calibration and decision-quality measures (e.g., Brier/ECE and MCC) provide practical separation even when F1 is similar, which matters for stable real-time alerting. In field deployments, practitioners should calibrate both the band and the alert threshold to local soil, crop, and sensor conditions.

Figure 2 presents ROC curves for all models and shows strong ranking separation, with curves clustered near the upper-left region. However, under skewed class distributions, ROC curves alone can be insufficient to summarize operating behavior; complementary views such as PR curves and confusion-sensitive metrics remain important [3], [5].

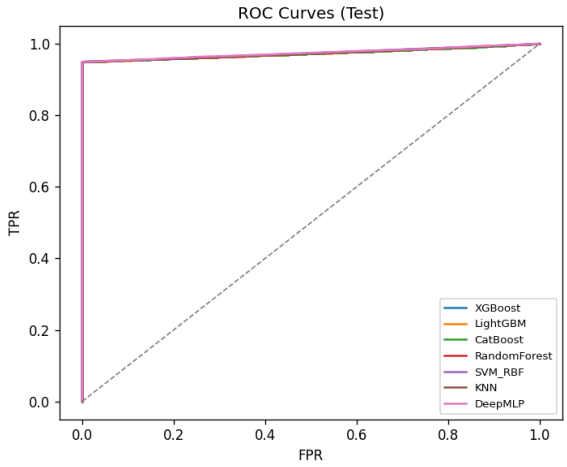


Figure 2. ROC curves for all evaluated models.

Figure 3 shows the precision–recall (PR) curves. While PR-AUC is often emphasized for rare-event detection [3], [5], it remains a useful precision/recall summary here as well; note that the baseline precision level equals the positive prevalence (~0.95), so near-ceiling PR-AUC should be interpreted relative to this high-prevalence baseline rather than as a rare-event indicator.

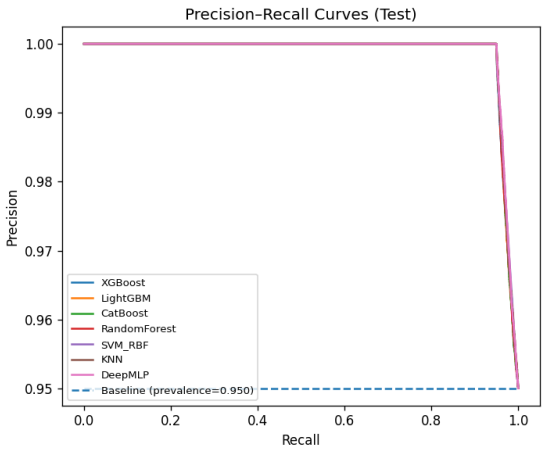


Figure 3. Precision–recall curves for all evaluated models.

Figure 4 reports calibration via reliability diagrams (uniform-bin) and supports the numeric calibration diagnostics: DeepMLP (ECE=0.0011, Brier=0.0244) and SVM_RBF (ECE=0.0016, Brier=0.0245) are the most reliable probabilistic predictors, while LightGBM attains the top F1 but exhibits higher calibration error (ECE=0.0234, Brier=0.0321). This distinction matters for real-time alerting because calibrated probabilities better support cost-aware threshold selection and stable alarm rates [4].

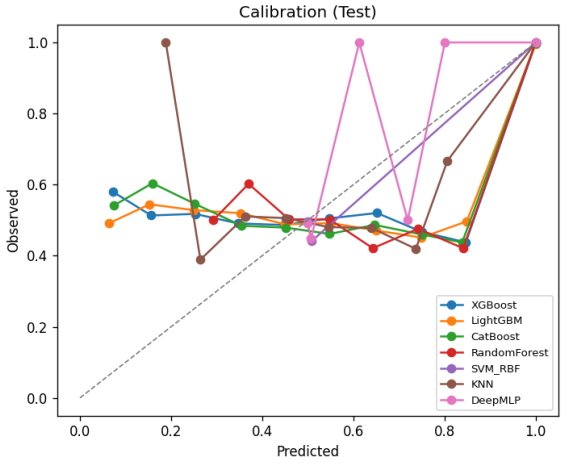


Figure 4. Reliability diagrams (calibration curves) comparing probability calibration.

Figure 5 summarizes the Matthews Correlation Coefficient (MCC) across the validation-tuned operating points. Despite similar F1 values, MCC can diverge because it penalizes confusion-matrix imbalance more strongly than F1, providing an additional check under skewed prevalence [5]. Notably, under F1-optimized thresholds, several models converge to an extreme high-recall operating point that flags almost all samples as anomalous (i.e., near-zero true negatives), yielding near-ceiling Recall and high F1 while producing MCC $\approx$ 0. In contrast, models that retain non-trivial true negatives at the tuned operating point achieve higher MCC, illustrating that threshold selection should be guided by deployment costs and desired alarm behavior rather than F1 alone under high positive prevalence.

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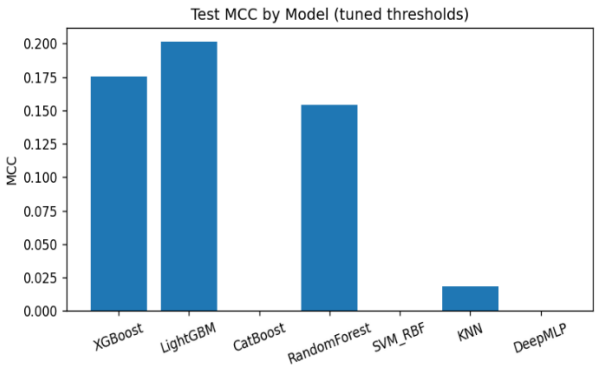


Figure 5. MCC comparison at the validation-tuned operating point.

Figure 6 visualizes pairwise McNemar tests on the tuned operating points. After Bonferroni correction (**α≈0.00238** for 21 pairwise tests), no significant disagreement pairs are detected, reinforcing that the observed differences are small relative to sampling variability at these operating points.

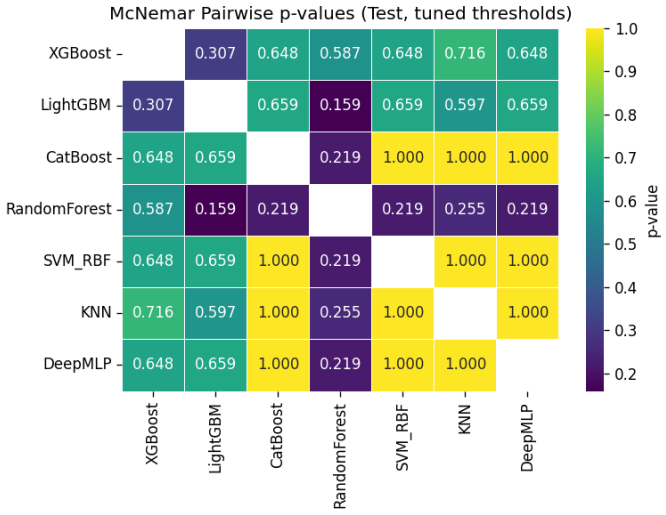


Figure 6. Pairwise McNemar-test significance matrix after correction.

Test-set bootstrap (N=200) yields narrow 95% intervals for F1 across models, e.g., LightGBM 0.97449 [0.97258, 0.97625], DeepMLP 0.97435 [0.97274, 0.97611], SVM_RBF 0.97434 [0.97256, 0.97625], CatBoost 0.97438 [0.97253, 0.97590], KNN 0.97428 [0.97242, 0.97602], XGBoost 0.97405 [0.97231, 0.97564], RandomForest 0.97383 [0.97187, 0.97564].

For deployment, thresholds should be selected by cost-aware validation and local prevalence; additionally, the "normal" moisture band should be aligned with FC/PWP for the target soil texture to ensure agronomic meaning and sensor consistency in field use [14]–[18].

VII. Conclusion

A leakage-safe benchmark with validation-tuned thresholds compares seven machine-learning models for real-time soil-moisture anomaly detection on a 100,000-record public dataset. All models yield high and tightly clustered F1 with near-ceiling PR-AUC. Calibration metrics and MCC provide additional discrimination under skew, guiding operating-point selection for field use. The artifact provides a reproducible accuracy reference for precision-agriculture monitoring and highlights that similar headline F1 values can mask meaningful differences in probability reliability and confusion balance.

Limitations and future directions. This benchmark is derived from a tabular public dataset and a fixed moisture-band proxy label; therefore, temporal drift and cross-season stability cannot be fully assessed without longitudinal field deployments or timestamped multi-season data. Future work will examine (i) distribution shift under cross-season/cross-site validation when such data are available, (ii) semi-supervised or self-supervised anomaly detection to reduce reliance on heuristic labels, (iii) dynamic thresholds that adapt to changing environmental regimes and target alarm rates, and (iv) multimodal sensing (e.g., temperature, EC, weather, irrigation context) to improve robustness beyond a single moisture-band proxy.

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